Entity Class Diagram

3.1 Recommendations and parking data

UC-01 to UC-03 | Anonymous search, explained results, carpark details and navigation destination

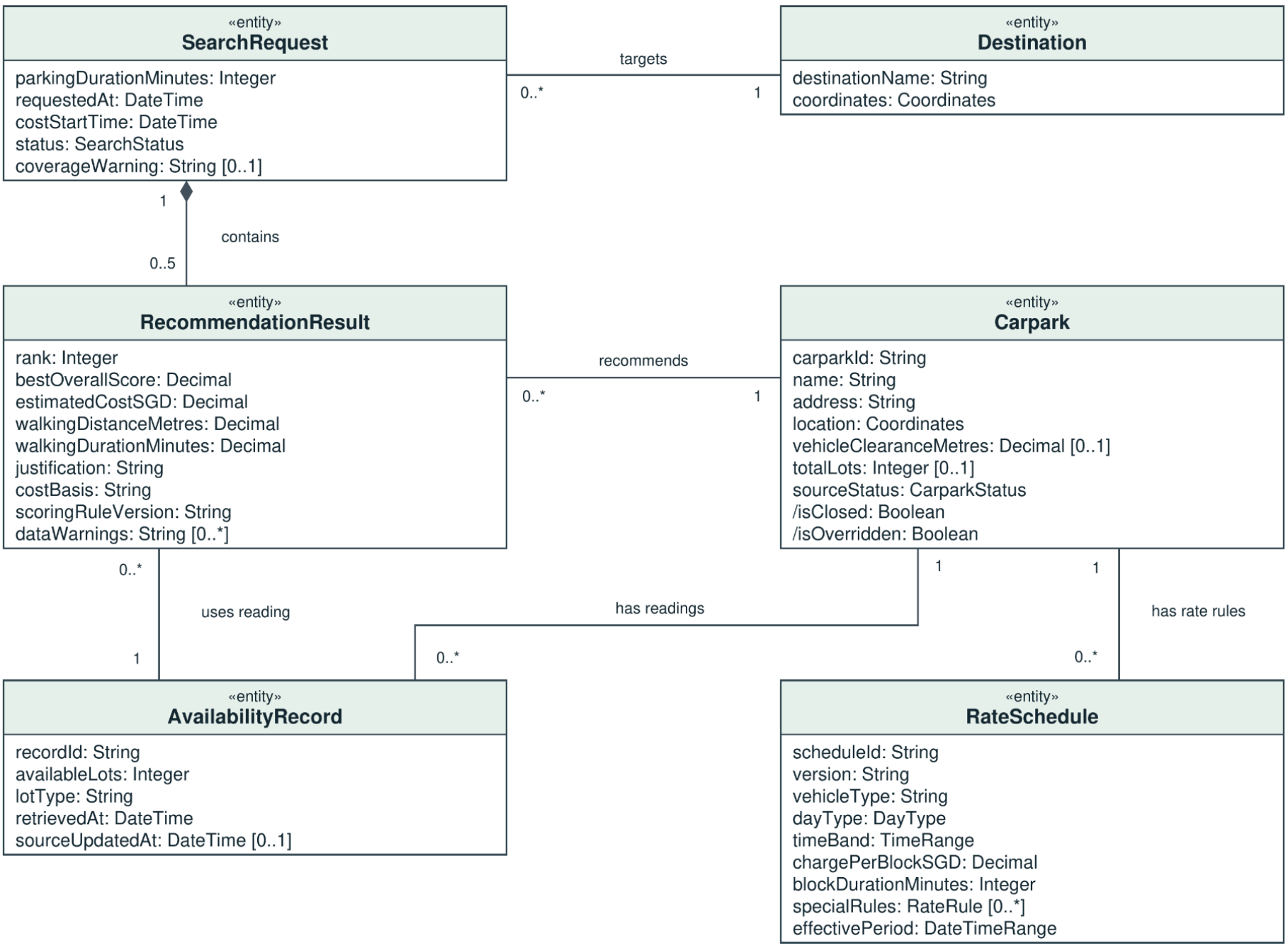


Figure 3.1. A SearchRequest targets one Destination and contains zero to five RecommendationResult objects. Each result recommends one Carpark and references the exact AvailabilityRecord used for its score. A carpark may have multiple availability readings and applicable rate rules over time.

Parking duration belongs to the request because different searches for the same destination can have different durations. The result holds the estimated charge, walking measurements, Best Overall score, explanation and warnings for that particular request. A selected carpark is an existing Carpark in the active results, rather than a separate subclass. Its name and location support the external-navigation handoff in UC-03.

The zero lower bound on readings and rate rules represents incomplete source data. It does not make an incomplete carpark eligible: UC-01 requires sufficiently current availability, valid rates and a walking measurement before a result can be generated. A completed successful search has one to five results; an unsuccessful search has none. Results have distinct ranks from 1 to the number returned, with rank 1 being Best Overall.

3.2 Reports, authentication and overrides

UC-04 to UC-07 | Anonymous submissions, operator review, session lifetime and reversible corrections

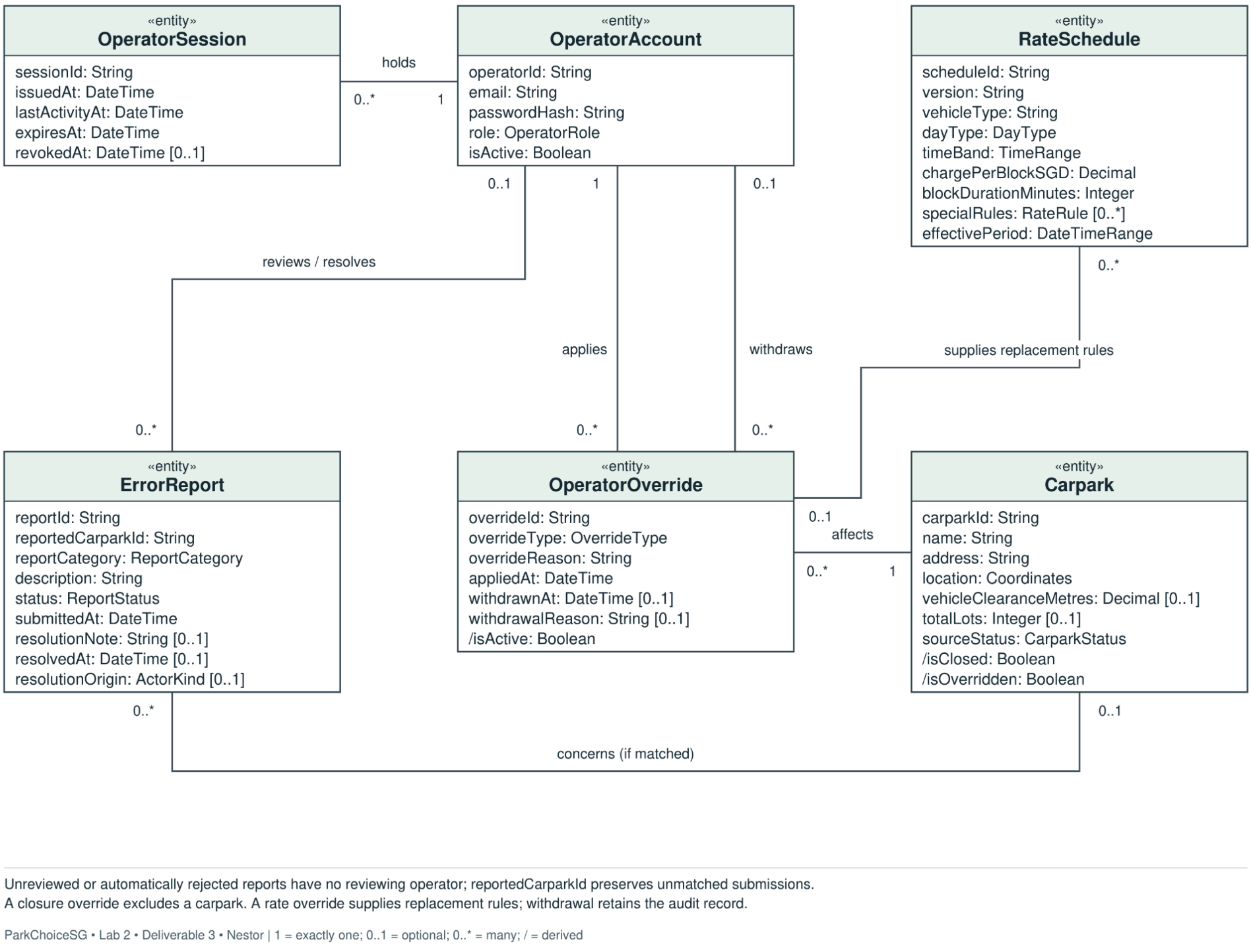


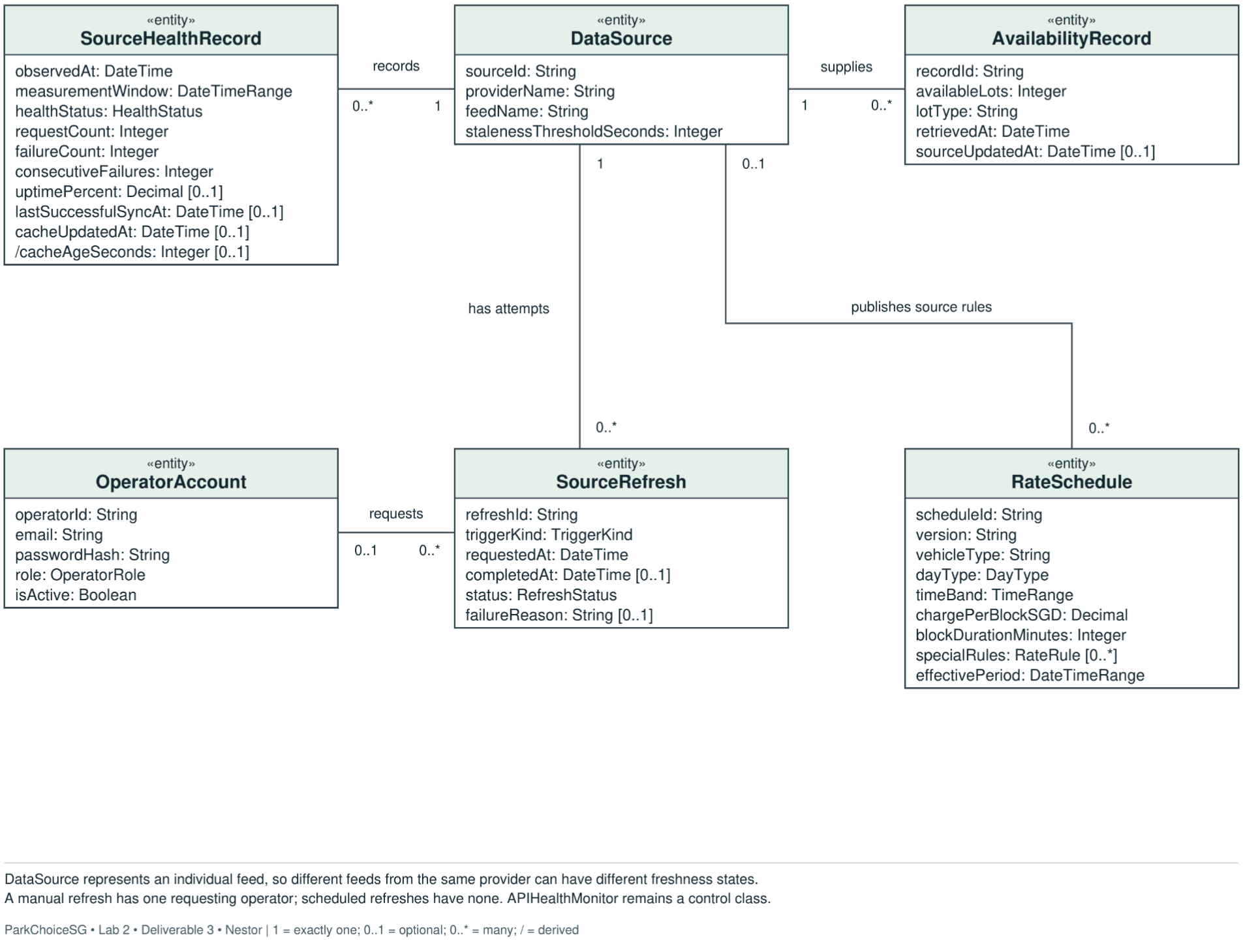
Figure 3.2. ErrorReport stores an anonymous submission and its eventual resolution. OperatorAccount holds administrative identity and is associated with its OperatorSession records, reviewed reports, and applied or withdrawn OperatorOverride records. Each override affects exactly one carpark; a rate override supplies replacement RateSchedule rules.

A report can exist without a reviewing operator. The optional association also supports automatic rejection when the submitted carpark identifier cannot be matched, as described in UC-06.AC.1. reportedCarparkId preserves the originally submitted identifier even when no Carpark association exists. Operator resolutions record the reviewing account, mandatory remarks, resolution time and outcome. Automatically rejected reports record a system origin, reason and resolution time.

An override records its author, reason and application time. Withdrawal records the withdrawing operator, time and reason while preserving the original record. Closure and rate correction are distinct override types: a closure excludes the carpark, while a rate correction replaces its applicable charging rules. Source refreshes must preserve active overrides. Carpark.isOverridden, Carpark.isClosed and OperatorOverride.isActive are derived from the source status and override records rather than independently editable flags.

3.3 Source health and refresh records

UC-08 and UC-09 | Per-feed freshness, operational measurements and manual or scheduled refresh attempts



DataSource represents an individual feed, so different feeds from the same provider can have different freshness states.
A manual refresh has one requesting operator; scheduled refreshes have none. APIHealthMonitor remains a control class.

ParkChoiceSG • Lab 2 • Deliverable 3 • Nestor | 1 = exactly one; 0..1 = optional; 0..* = many; / = derived

Figure 3.3. DataSource identifies an individual external feed. SourceHealthRecord contains the measured health and freshness information shown to operators. SourceRefresh records each refresh attempt and its outcome. Manual attempts reference the requesting operator; scheduled attempts do not.

Availability records identify their supplying source. Source-provided rate schedules identify their source; replacement rules from manual overrides are linked through OperatorOverride in Figure 3.2. A provider may expose multiple feeds with different refresh frequencies, so the model monitors feeds individually. APIHealthMonitor collects and interprets these records as a control class in Deliverable 4; it is not an entity.

3.4 Entity responsibilities and use-case traceability

Entity	Responsibility	Use cases
SearchRequest	Captures duration, calculation time basis and search outcome	UC-01
Destination	Describes the resolved place being visited	UC-01, UC-02
RecommendationResult	Retains ranked factors, estimated cost, explanation and evidence used	UC-01, UC-02
Carpark	Describes parking location, restrictions and effective operational status	UC-01–UC-04, UC-06, UC-07
AvailabilityRecord	Records a lot reading with source and retrieval time	UC-01, UC-02, UC-08, UC-09

RateSchedule	Describes versioned charging rules for a day/time period	UC-01, UC-02, UC-07
ErrorReport	Records an anonymous issue and its resolution	UC-04, UC-06
OperatorAccount	Identifies an authorised administrator	UC-05 - UC-09
OperatorSession	Represents the lifetime and expiry of administrative access	UC-05; access precondition for UC-06 - UC-09
OperatorOverride	Preserves a manual correction and its application/withdrawal audit information	UC-06, UC-07
DataSource	Identifies each external feed and its freshness threshold	UC-08, UC-09; evidence provenance in UC-01/UC-02
SourceHealthRecord	Stores a timed observation of source health and cache freshness	UC-08
SourceRefresh	Records a scheduled or manually requested refresh and outcome	UC-09, UC-08 alternative flow